Trouvaille: AI-Powered Outfit Recommendation
Deliverable 1 – Problem Definition & Data Foundations

Group Members Romesha Afzaal 24L-3001 Aliza Nadeem 24L-3024

1 Phase 1: Problem Definition

1.1 Problem Statement

Online fashion shopping is time-consuming and fragmented. A shopper preparing for an event, such as a wedding, concert, presentation or office day, must search for each clothing item separately, keep a budget in mind, and guess whether the pieces will look good together. Existing platforms rely on keyword search and category filters, which work poorly for requests like "a black outfit under Rs. 15,000 for a concert", because this single sentence combines colour, budget, occasion and multiple garments.

A second pain point is that most platforms ignore the individual. Recommendations are rarely adjusted for skin tone or body type, so shoppers often receive items that do not suit them, and rely on product ratings alone as a proxy for quality. Finally, no mainstream tool checks whether a complete outfit is appropriate for the type of event. The result is wasted time, over-budget purchases, returns, and low confidence in online buying.

Trouvaille addresses this by letting users describe what they need in plain language and returning a complete, budget-aware, occasion-appropriate outfit built from real products.

1.2 Target Audience

The primary users are online shoppers in Pakistan, particularly students and young professionals, who buy clothing for specific events on a limited budget and want quick, confident styling decisions. Secondary users include shoppers who find fashion choices overwhelming, and people who are underserved by generic recommendations because of their body type or skin tone.

From a research perspective, the project targets a gap between existing work: most recommender systems rank single items, whereas Trouvaille combines (i) natural-language query understanding, (ii) inclusive re-ranking by skin tone, body type and ratings, (iii) budget-constrained outfit composition, and (iv) an occasion-appropriateness check, all in one pipeline.

1.3 AI Justification

A traditional rule-based system is not suitable for this problem for three reasons.

Free-form language understanding. Users phrase requests in countless ways ("something black for a concert under 15k", "formal look for my cousin's wedding, size M"). Hand-written rules or regular expressions cannot cover this variety. A Large Language Model can reliably extract structured attributes such as budget, colour, occasion and size from unstructured text.

Semantic product matching. Keyword matching fails when a query and a product description use different words for the same idea (e.g. "edgy" or "smart casual"). Sentence embeddings represent queries and products as vectors, so products can be retrieved by meaning rather than exact wording, and the system generalizes to products it has never seen.

Subjective, context-dependent judgment. Deciding whether an outfit suits a wedding versus an office, or whether items complement each other, cannot be reduced to a fixed set of if–else rules because the criteria are subjective and overlap. Learned models capture these soft relationships from data and scale as the product catalogue grows, whereas a rule base would need constant manual maintenance.

Therefore, machine learning and deep learning are necessary for the language understanding, retrieval and compatibility components of Trouvaille, while deterministic logic is used only where exact constraints apply, such as enforcing the budget.

2 Phase 2: Data Acquisition & Preprocessing

2.1 Data Collection

Product data was collected from the online stores of ten Pakistani clothing brands (**Outfitters, Ethnc, Bonanza Satrangi, Lama, Lulusar, Faiza Saqlain, Hussain Rehar, Saira Shakira, J. (Junaid Jamshed) and Vanya**) using the script `scrape_products.py`. The script reads each store's public product feed (`/products.json`), checks the store's `robots.txt` before requesting data, waits between requests, retries failed requests, and records every step in a log file (`scrape_log.txt`), which serves as evidence of collection. For each product we store the title, type, tags, description, price, compare-at price, available sizes and colors, stock status, image URL and product URL. In total, **40,537** raw products were collected on **24 October 2026**. The number of raw products per brand was: J. 12,500; Bonanza Satrangi 10,089; Ethnc 4,960; Outfitters 3,885; Lulusar 2,781; Hussain Rehar 2,162; Lama 1,616; Faiza Saqlain 992; Vanya 834; and Saira Shakira 718. One additional store (Sapphire) was attempted but its feed returned a 404 error, so it was excluded. The scraper stops after a maximum of 50 pages per store, and this limit was reached for J., so its catalog may be larger than the 12,500 products collected.

2.2 Data Cleaning

The script `clean_data.py` performs the following steps (counts are in `cleaning_report.txt`):

- **Duplicates:** products repeated across pages or collections were removed using brand and product handle, and then brand, title and price (**3,403** removed).
- **Missing values:** products without a title or a valid price were dropped (**240**). Missing colors were filled by extracting the color from the title (remaining unknown colors are labeled `unknown`); missing sizes are labeled `UNKNOWN`; missing categories were inferred from the product type and title keywords, and products that could not be assigned a clothing category were dropped (**5,688**). The raw data had missing values mainly in `description` (7,401 products), `colors` (9,127), `compare_at_price` (7,277, meaning the product is not on sale), `sizes` (5,737) and `product_type` (838).
- **Standardization:** HTML was stripped from descriptions, sizes were mapped to a common scheme (e.g. `XXL` to `2XL`), and colors were normalized to a fixed vocabulary.

- **Scope:** children's products were removed because the application targets adult outfits (**4,978**).
- **Outliers:** extreme prices were removed using the interquartile-range rule on log-price within each category (**1,261** removed), which prevents data-entry errors and luxury items from distorting the budget-based search.

The final cleaned dataset contains **24,967** products (61.6% of the raw data) from **10** brands. However, the brand coverage is uneven: Faiza Saqlain and Saira Shakira keep only 4 and 3 products respectively, because their products carry first names as titles (e.g. "Seda") and have no category keyword, or have placeholder prices of zero, so they were removed by the category and price rules. In practice the dataset is built from eight brands.

2.3 Feature Strategy

The script `preprocess_features.py` prepares features for the later models.

- **Scaling:** price is min–max scaled to $[0, 1]$; log-price is standardized (z-score) because prices are strongly right-skewed; discount percentage is min–max scaled.
- **Encoding:** category, gender, grouped color (top 12 colors plus `other`), brand, color undertone, fit type and formality are one-hot encoded. Occasions and sizes can take several values per product, so they are multi-hot encoded.
- **Text features:** each product has a `text_for_embedding` field combining its title, attributes and description, to be encoded with sentence embeddings in the next phase.

Derived labels for later components. The Vibe Check Engine and the inclusive filter need information that stores do not provide, so we derived heuristic labels: (i) occasion labels (wedding, office, presentation, concert, casual) and a formality level from keyword rules; (ii) a fit type (fitted, relaxed, regular) and stretch flag from keywords; (iii) a color-undertone group (warm, cool, neutral) from a color-analysis rule, with the skin undertones each group is assumed to suit; and (iv) a plus-size indicator (2XL or larger offered) as a measurable proxy for body-type inclusivity. These labels are approximations and not ground truth, and will be validated and refined in later phases. Empty columns for manually assigned skin-tone and body-type labels are reserved for that purpose. Customer ratings are not available from the store feeds and are left for a later phase.

2.4 Exploratory Data Analysis

The catalog is dominated by sets (50.8% of products), followed by tops (20.4%) and bottoms (14.3%), while footwear (3.2%) and outerwear (2.1%) are rare; women's products make up 57.2% of the data, men's 22.9% and unisex 19.9%, and two brands, Bonanza Satrangi and J., supply 68.7% of all products. The occasion labels are multi-label and skewed toward casual (68.9%), and office (52.9%) and presentation (53.0%) have almost identical counts because their keyword lists overlap, so these two labels carry little independent information; wedding (42.2%) is probably inflated by broad keywords such as "embroidered" or "chiffon". Prices are strongly right-skewed (median PKR 4,000, mean 4,934, maximum 65,000, skewness 2.6), which justifies the log transform, and outerwear is the most expensive category (median PKR 7,693) while accessories and footwear are the cheapest. Price is moderately negatively correlated with discount percentage (-0.54), meaning cheaper items tend to be discounted more heavily, and 66% of products carry a discount, so the data reflects a sale period and the discount feature may not generalize to other times. Size inclusivity is limited:

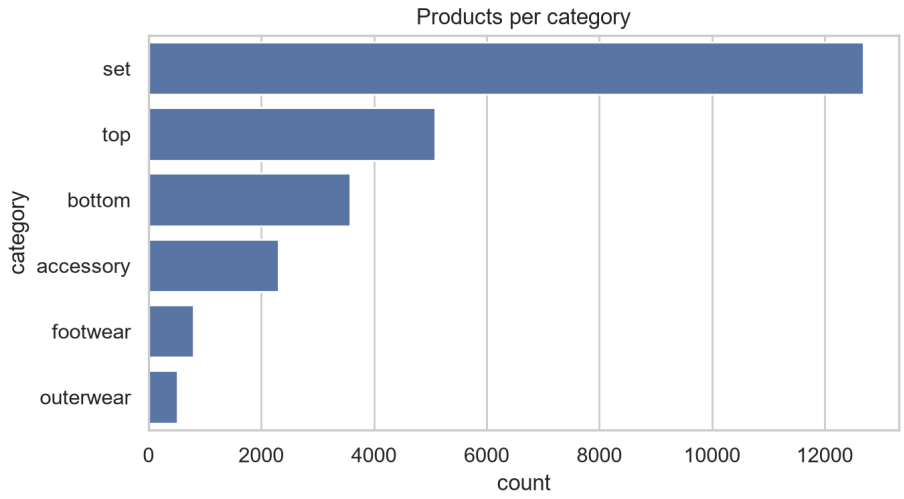


Figure 1: Number of products per category.

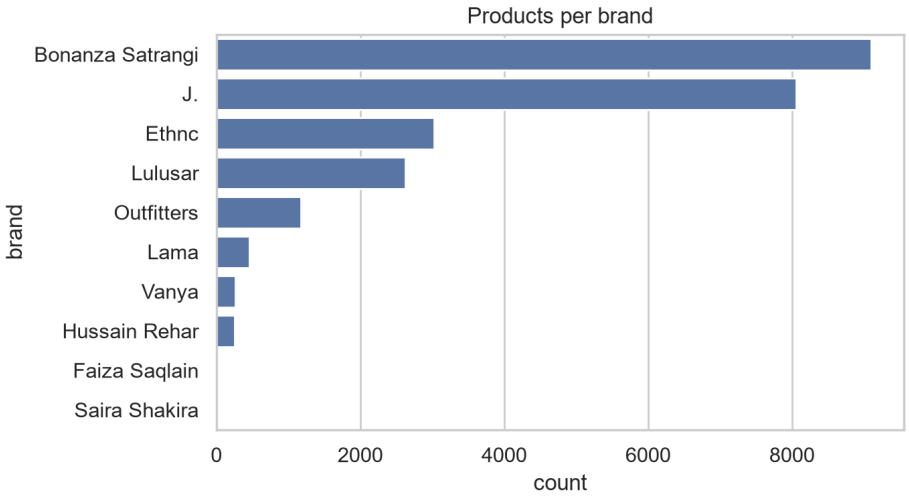


Figure 2: Number of products per brand.

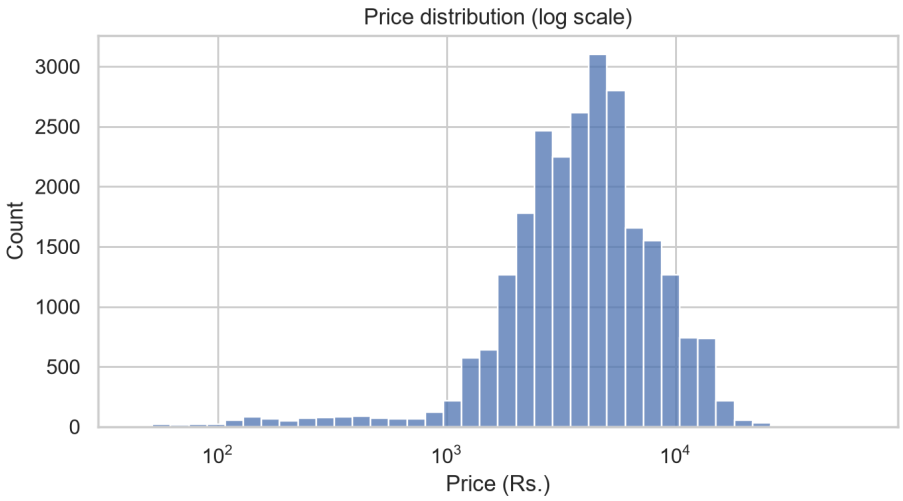


Figure 3: Price distribution (log scale).

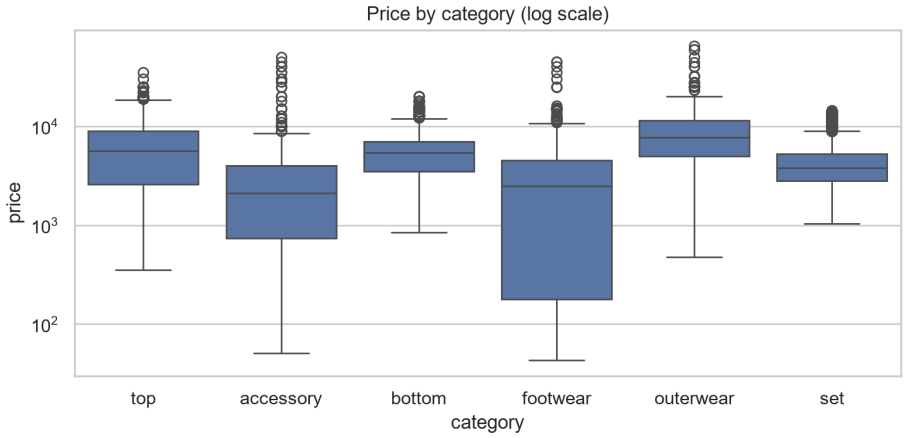


Figure 4: Price by category (log scale).

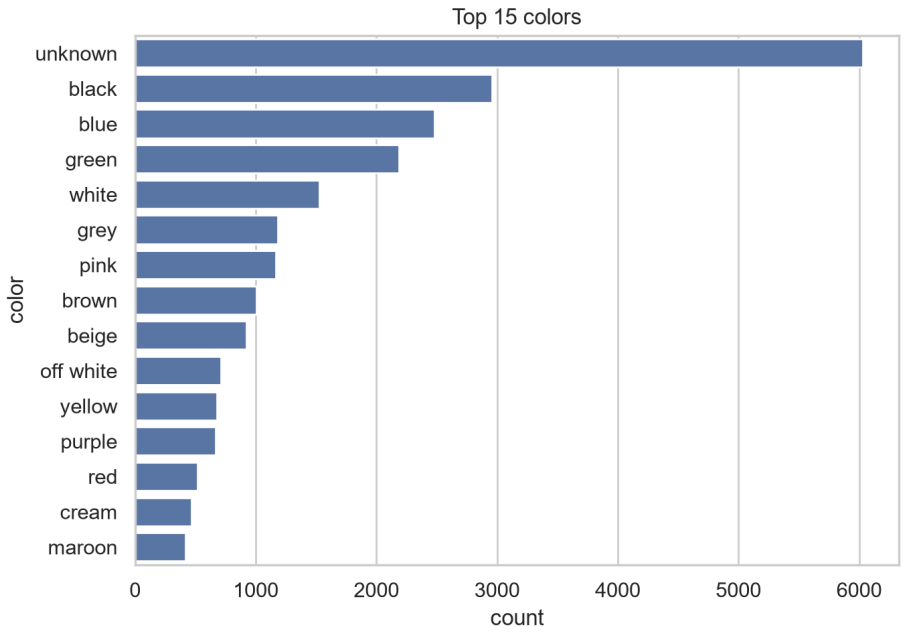


Figure 5: Most common colors.

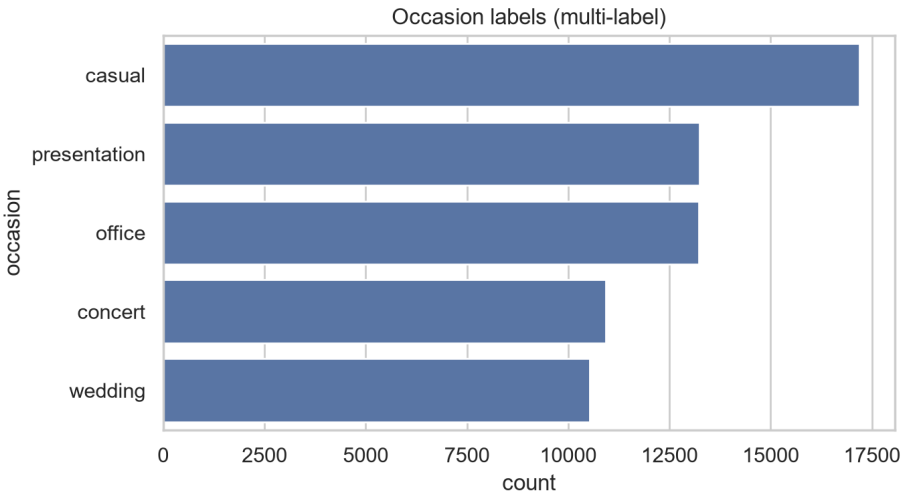


Figure 6: Distribution of heuristic occasion labels.

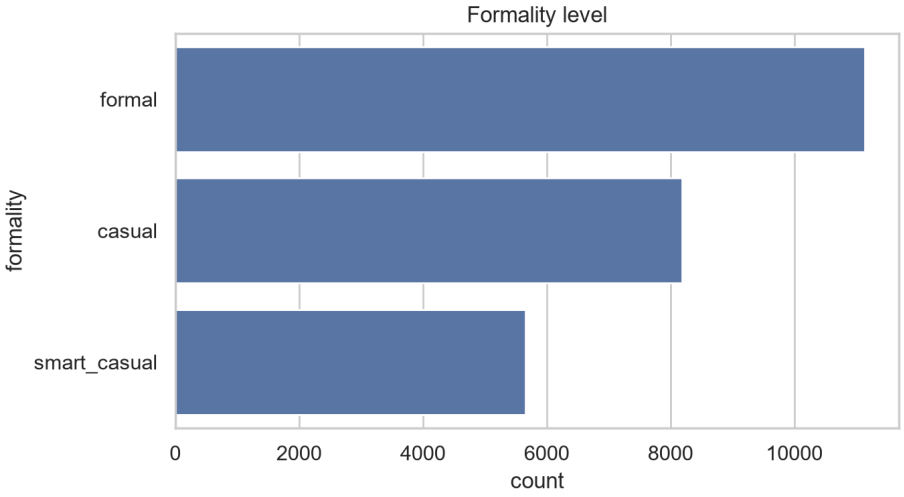


Figure 7: Distribution of formality levels.

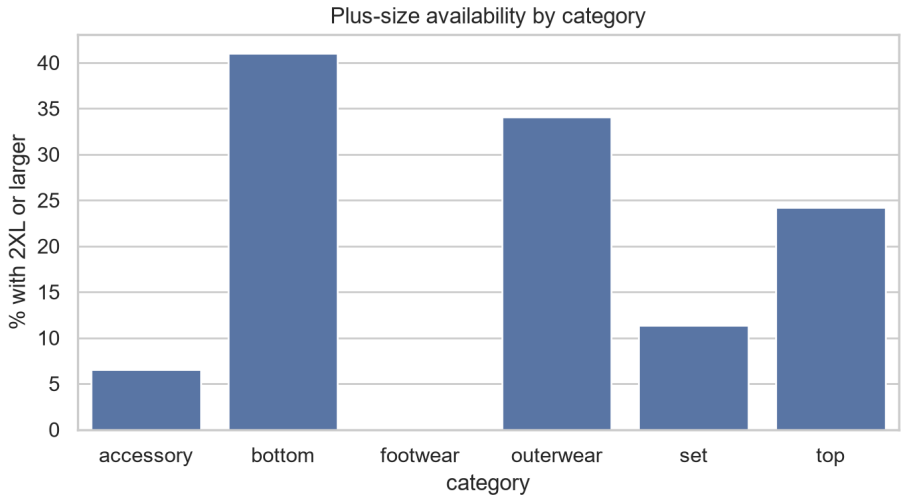


Figure 8: Share of products offering 2XL or larger, by category.

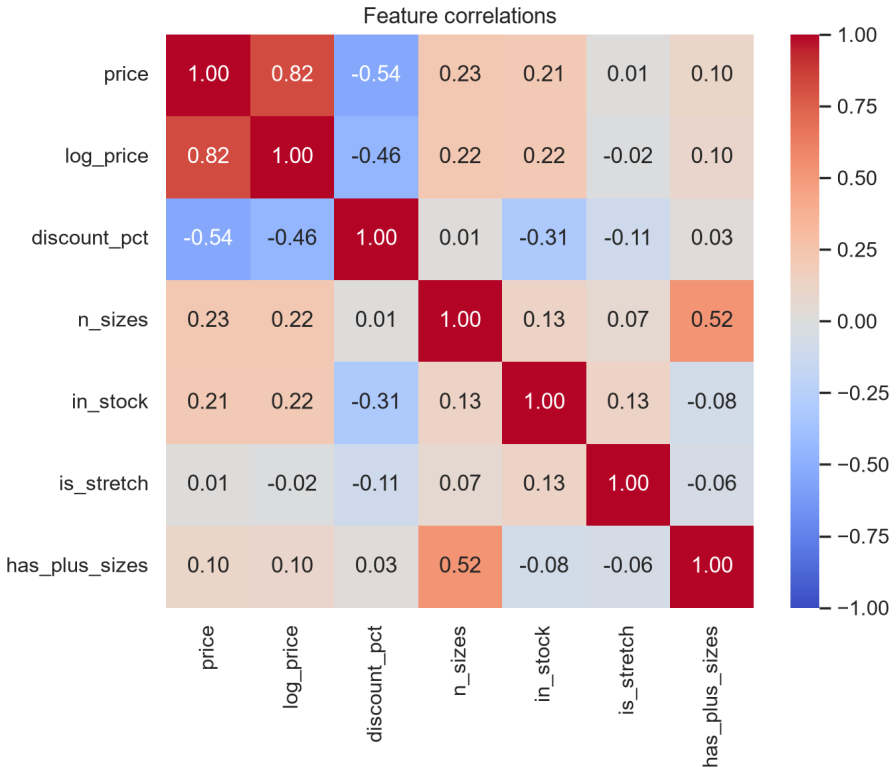


Figure 9: Correlation between numeric features.

only 17.9% of products (4,468) offer 2XL or larger, none of the footwear does, and about 91% of all plus-size products come from a single brand (J.), so the inclusive filter will have few options for most women's categories (14.4% of women's products offer plus sizes, against 42.2% of men's). For modeling, this means the data should be split in a way that is stratified by category and brand, the imbalance handled with class weights or resampling, the redundant features removed (`price` and `log_price` correlate at 0.82, and `has_plus_sizes` correlates at 0.52 with the number of sizes from which it is derived), and the heuristic labels evaluated carefully since they come from the same keywords as the text the models will see.

2.5 Limitations

Occasion, fit and undertone labels are keyword-based heuristics; ratings and skin-tone/body-type labels are not yet available; and the data covers only the selected brands at one point in time. In addition, the sample is concentrated in two brands, the luxury and bridal segment is almost absent after cleaning (Faiza Saqlain and Saira Shakira), the J. catalog was truncated at 50 pages, and 24.1% of products have an unknown color. Some prices, notably for Lama (median PKR 178), look unusually low compared with other brands and may use a different price unit in the feed; this should be checked before price is used as a model feature.